

TERRALINK PHASE-5

Testing, Performance & Reliability Evaluation

DYNAMIC TERRALINK

Emergency Communication System using LoRa, GPS and Relay-Based Communication

REPORT SCOPE	Functional testing • Performance evaluation • Test cases • Error analysis • Debugging • Power • Reliability • Future improvements
SYSTEM PATH	Trigger → GPS → SOS Packet → LoRa Transmission → Relay → Rescue Station → Status
EVALUATION	ESP32 sender • LoRa mesh forwarding • GPS • OLED status • relay nodes • rescue/base-station receiver

Important evaluation note

Numerical performance and power values are not assumed or fabricated. Final numerical claims must be supported by recorded physical measurements.

1. FUNCTIONAL TESTING

Functional testing verifies the complete TerraLink emergency communication sequence from user trigger through final status.

Function	Expected Behaviour	Validation Focus
SOS Trigger	Touch input detects the emergency request.	TTP223 response and wake-up.
GPS	Location is acquired when required.	Latitude and longitude availability.
SOS Packet	Emergency information is assembled.	Node ID, SOS, GPS, battery, packet ID and TTL.
LoRa Transmission	Packet is transmitted through the LoRa link.	Sender transmission.
Relay Forwarding	Valid packet is received and forwarded.	Multi-hop operation.
Rescue Station	Final packet is received.	End-to-end delivery.
OLED / Status	User receives local status information.	Awake / alert transmitted display.

Testing Principle

Each functional block is verified independently and then as part of the complete emergency path. Unsupported numerical results are not fabricated; measured values must come from the physical experiment.

Functional Validation Flow

Touch Input → ESP32 Processing → GPS Data → SOS Packet → LoRa TX → Relay Forwarding → Rescue/Base Station → Status

Expected Outcome

A valid emergency event should be detected, the required packet information assembled, the packet transmitted through the intended LoRa path, and the final receiver should obtain the emergency information.

2. PERFORMANCE EVALUATION

Performance evaluation measures how effectively TerraLink delivers emergency packets across the LoRa mesh and how quickly the system responds.

Parameter	What is Evaluated	Result / Record
Packet Delivery	Successful packets reaching the receiver.	Record measured value.
End-to-End Delay	Time from SOS generation to final reception.	Record measured value.
Communication Range	Maximum practical communication distance.	Record measured distance.
Multi-Hop Forwarding	Delivery through Relay 1 and Relay 2.	Physical validation.
Duplicate Handling	Behaviour when an existing packet is received again.	Validation result.
GPS Response	Location acquisition time and correctness.	Record measured result.
RSSI / SNR	Signal quality at each communication hop.	Record observed values.

Performance Interpretation

- High packet delivery ratio indicates that the communication path is functioning consistently.
- Low and predictable end-to-end delay is important because emergency information must reach the rescue station promptly.
- Successful multi-hop forwarding demonstrates that TerraLink can extend communication beyond a direct link.
- Effective duplicate rejection prevents repeated packets from creating repeated alerts or unnecessary forwarding.
- Fast recovery after a node or link failure improves the availability of the communication network.

Important: Actual numerical performance values must be entered from recorded experimental measurements.

3. TEST CASES — TC-01 TO TC-09

Large-format validation matrix for clear review and physical testing.

ID	TEST CASE	EXPECTED RESULT	STATUS
TC-01	Power ON / Boot	ESP32 starts correctly and initializes required modules.	Implemented / Validate
TC-02	SOS Trigger	TTP223 touch activates emergency request.	Physically Test
TC-03	Deep-Sleep Entry	Node enters low-power state when idle.	Physically Test
TC-04	Wake from SOS	Touch event wakes the sender from sleep.	Physically Test
TC-05	OLED Wake Display	OLED shows TerraLink wake/status information.	Physically Test
TC-06	GPS Acquisition	GPS obtains valid latitude and longitude when required.	Physically Test
TC-07	SOS Packet Generation	Emergency packet contains required information.	Implemented / Validate
TC-08	Packet ID Uniqueness	New emergency events use distinct packet IDs.	Validation / Planned
TC-09	LoRa Transmission	Sender transmits the emergency packet.	Physically Test

3. TEST CASES — TC-10 TO TC-18

Large-format validation matrix for clear review and physical testing.

ID	TEST CASE	EXPECTED RESULT	STATUS
TC-10	Relay 1 Reception	Receives valid sender packet.	Physically Test
TC-11	Relay 1 Forwarding	Forwards valid packet.	Physically Test
TC-12	Relay 2 Reception	Receives forwarded packet.	Physically Test
TC-13	Relay 2 Forwarding	Forwards packet to receiver.	Physically Test
TC-14	Receiver Reception	Base/rescue station receives emergency packet.	Physically Test
TC-15	End-to-End Alert	SOS reaches rescue station through mesh.	Physically Test
TC-16	Duplicate Detection	Repeated copy is discarded.	Validation / Planned
TC-17	TTL Handling	TTL is reduced on forwarding and expired packets stop.	Validation / Planned
TC-18	Invalid Packet Handling	Malformed/invalid packets are safely rejected.	Validation / Planned

3. TEST CASES — TC-19 TO TC-25

Large-format validation matrix for clear review and physical testing.

ID	TEST CASE	EXPECTED RESULT	STATUS
TC-19	RSSI/SNR Observation	Link quality is observed at each hop.	Physically Test
TC-20	Weak-Link Behaviour	Behaviour is checked under weak signal.	Validation / Planned
TC-21	Relay/Link Failure	Communication is checked when relay/link fails.	Validation / Planned
TC-22	Repeated SOS Reliability	Repeated emergency events are checked.	Physically Test
TC-23	Power Consumption	Current/power is compared across states.	Physically Test
TC-24	Restart Recovery	System operates normally after restart.	Physically Test
TC-25	Long-Duration Operation	Stable extended operation is checked.	Validation / Planned

4. ERROR ANALYSIS & 5. DEBUGGING ANALYSIS

The following development issues were identified during TerraLink development and linked to corrective debugging actions.

ERROR / ISSUE	IMPACT	CORRECTIVE ACTION
OLED not turning ON	No local status display.	Check I2C wiring, address, initialization and SH1106 configuration.
SOS not sending	Emergency packet was not transmitted.	Verify touch input, wake sequence, LoRa initialization and send routine.
GPS lock slow	Location unavailable immediately.	Allow acquisition time and activate GPS only when required.
Deep-sleep components stayed ON	Expected power saving was not achieved.	Control OLED/GPS/LoRa power states and verify sleep configuration.
Touch did not wake	Emergency wake event failed.	Verify wake-capable GPIO and touch signal path.
LoRa receive/forward issue	Mesh delivery was interrupted.	Check frequency, SPI pins, sync word, packet format and forwarding logic.
Duplicate packet issue	Repeated processing/forwarding could occur.	Use packet-ID tracking and duplicate rejection where implemented.

Debugging Method

- Reproduce the problem and confirm that the error is consistent.
- Identify the affected module: trigger, GPS, packet processing, LoRa, relay, display or power.
- Inspect inputs, trace processing with serial/debug messages, and inspect outputs.
- Apply only the relevant hardware, configuration or software correction.
- Retest the failed test case and perform regression testing.

Subsystem checks

OLED: I2C connections, initialization and SH1106. SOS: TTP223 input, wake sequence and send routine. GPS: serial connections and acquisition time. Deep Sleep: sleep entry, wake source and peripheral power state. LoRa: 433 MHz configuration, SPI pins, reset/DIO, sync word and packet flow. Mesh: Sender →

Relay 1 → Relay 2 → Receiver. Duplicate handling: packet-ID tracking.

6. POWER CONSUMPTION ANALYSIS

Power analysis evaluates the battery-oriented design of TerraLink. Deep sleep reduces idle activity, while SOS events activate the functions required for emergency communication.

STATE	POWER STRATEGY	MEASUREMENT TO RECORD
Deep Sleep	ESP32 and unnecessary peripherals kept in low-power state.	Sleep current.
Wake / SOS	Required functions become active.	Wake current.
GPS Active	GPS operates when location is required.	GPS current and acquisition duration.
LoRa TX	Radio is active during packet transmission.	TX current and duration.
LoRa RX / Relay	Radio receives and forwards emergency packets.	RX current and duration.
OLED	Used for local status; ideally off during sleep.	Display current.
Battery Operation	Compare duty cycle and operating states.	Estimated runtime from measured data.

Measurement Guidance

Actual current values should be obtained using a suitable current/power measurement instrument. Battery-life figures should be calculated from measured data.

Power calculation: $P = V \times I$, where P is electrical power in watts, V is voltage in volts and I is current in amperes.

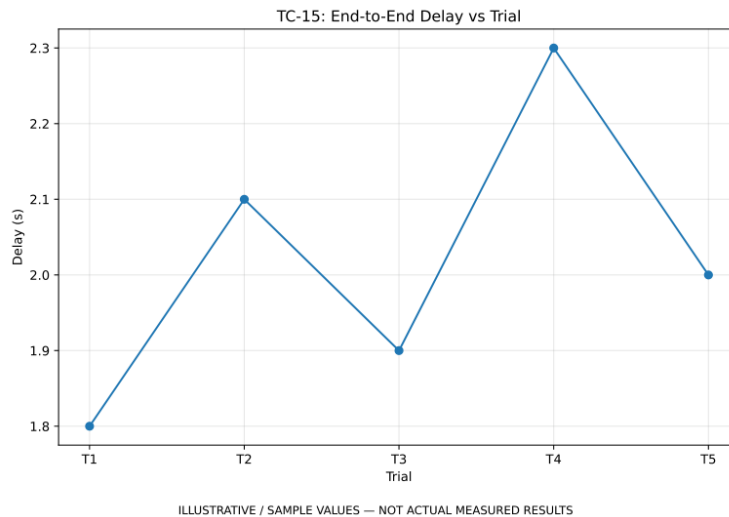
Do not use nominal module current values as the final project measurement. Use values measured from the actual TerraLink hardware.

7. ILLUSTRATION GRAPHS

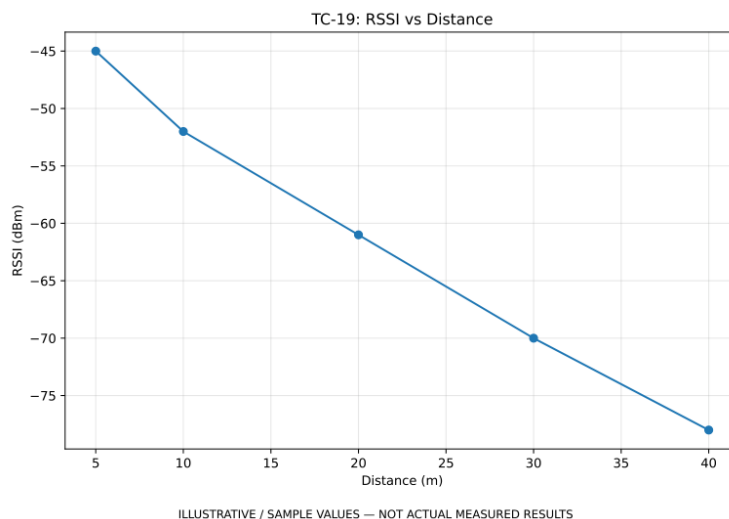
Supporting visuals for the Phase-5 evaluation

⚠ These graphs are explicitly illustrative/sample values, not actual measured results. Replace them with recorded experimental graphs before claiming numerical performance.

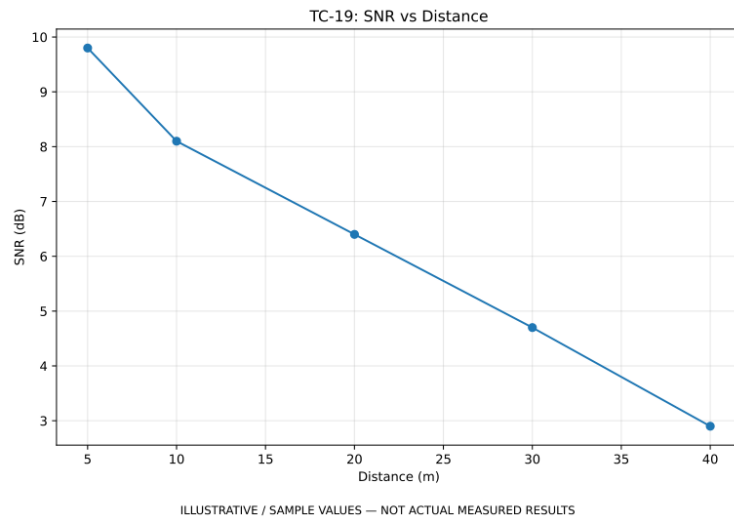
TC-15: End-to-End Delay vs Trial



TC-19: RSSI vs Distance



TC-19: SNR vs Distance

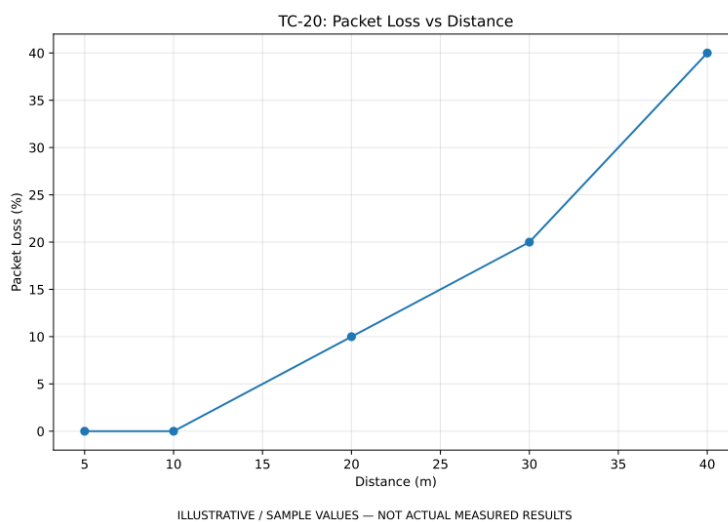


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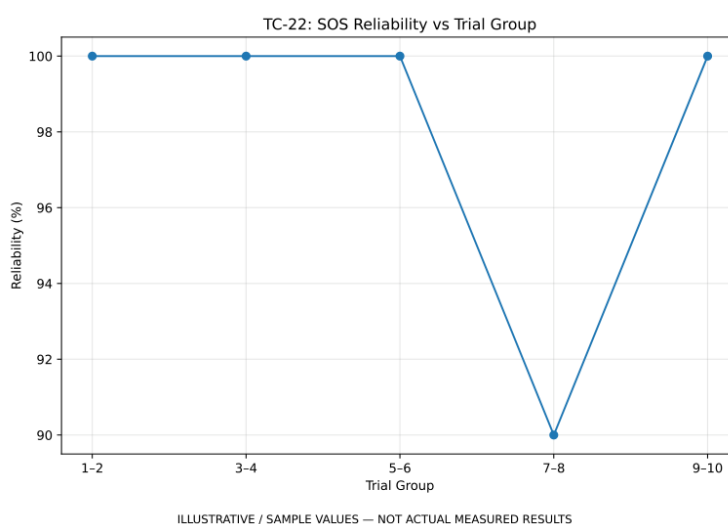
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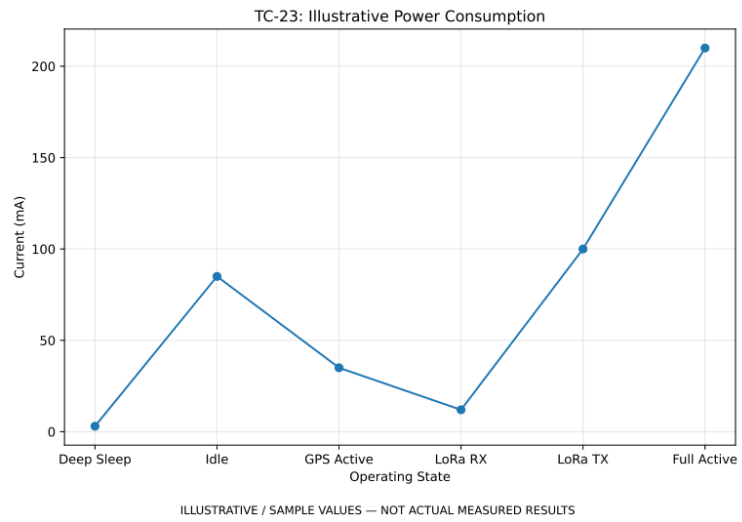
TC-20: Packet Loss vs Distance



TC-22: SOS Reliability vs Trial Group



TC-23: Illustrative Power Consumption



8. RELIABILITY TESTING & FUTURE IMPROVEMENTS

Reliability testing checks whether TerraLink continues to operate correctly across repeated alerts, multi-hop forwarding, weak links, relay interruptions and restarts.

TEST AREA	RELIABILITY CHECK	ACCEPTANCE FOCUS
Repeated SOS	Send multiple emergency events.	Consistent packet generation and transmission.
Multi-Hop Path	Sender → Relay 1 → Relay 2 → Receiver.	Successful forwarding.
Weak Signal	Observe operation under reduced signal quality.	Graceful communication behaviour.
Relay Interruption	Interrupt a relay or communication link.	Safe handling of the changed path.
Duplicate Packet	Repeat an existing packet.	Duplicate handling.
Restart Recovery	Restart the node after operation.	Normal recovery.
Long Duration	Operate for an extended period.	Stable operation without unexpected failure.

Future Improvements

- Dynamic relay selection: deploy portable relay nodes where additional coverage is required.
- Adaptive routing: use link quality, hop count, battery level and reliability for route selection.
- Acknowledgement mechanism: provide stronger confirmation that the emergency packet reached the rescue station.
- Improved duplicate handling: maintain packet history to prevent repeated processing.
- Battery optimization: further reduce idle consumption and optimize GPS/LoRa activation.
- Expanded reliability testing: validate weak links, relay failure and longer operating periods.
- Data logging: record RSSI, SNR, delay, delivery and power measurements for quantitative evaluation.

9. OVERALL TEST SUMMARY & CONCLUSION

Phase-5 provides the final structured validation layer for the TerraLink prototype. The testing process covers functional behaviour, communication performance, test-case execution, error identification, debugging, power consumption and reliability.

Reliability objective

The emergency alert should remain correctly formatted and processed throughout the intended communication path, while failure and duplicate conditions are handled safely.

Conclusion

TerraLink integrates SOS triggering, GPS location information, low-power ESP32 operation and relay-based LoRa multi-hop communication for emergency communication. The Phase-5 test plan covers functional behaviour, performance, error correction, power consumption and reliability.

Final numerical claims should be based only on recorded experimental measurements.

CATEGORY	FINAL OBSERVATION / RECORD
Functional testing	_____
Performance evaluation	_____
Test-case execution	_____
Error analysis	_____
Debugging verification	_____
Power consumption	_____
Reliability testing	_____

PHASE-5 COMPLETION STATEMENT

This report presents a structured evaluation framework for the TerraLink prototype, documenting what was implemented, what was tested, the issues encountered, the corrective debugging approach, the reliability checks and the next engineering improvements.